

Category Boundary Sorting Classroom Run Kit

TOK Cognitive Science Activity Bank • Knowledge and Language • 25-35 min

Category	Knowledge and Language	Time	25-35 min
Difficulty	Medium	ID	category-boundary-sorting

Big idea: Categories are made for purposes.

One-Minute Teacher Brief

Students investigate how categories depend on criteria and purposes.

Student Prompt

Inspect Category Boundary Sorting Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Category Boundary Sorting Stimulus A	Student-facing stimulus 1: Cards such as game, sport, art, science, medicine, belief, evidence, and opinion. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Category Boundary Sorting Stimulus B	Student-facing stimulus 2: Borderline cases that force students to define category rules. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Use this for comparison, a second round, or a partner challenge.
Category Boundary Sorting Reveal / Twist	Student-facing stimulus 3: A rule revision round after students encounter exceptions. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Teacher reveal: connect the changed response to the big idea — Categories are made for purposes.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Category Boundary Sorting Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Category Boundary Sorting Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Do categories describe reality or organise it?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Language.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how categories are made for purposes.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Do categories describe reality or organise it?

Setup Checklist

- Prepare a starter stimulus using: Cards such as game, sport, art, science, medicine, belief, evidence, and opinion.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- Cards such as game, sport, art, science, medicine, belief, evidence, and opinion.
- Borderline cases that force students to define category rules.
- A rule revision round after students encounter exceptions.

Resources To Use

- Resource pack: <resources/category-boundary-sorting/index.html>
- Card deck CSV: <resources/category-boundary-sorting/cards.csv>
- Visual prompt: <resources/category-boundary-sorting/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Groups sort prompts into yes, no and depends.
2. For each "depends", they must state the criterion.
3. Groups trade criteria and test them on borderline cases.
4. Discuss why different communities define the same term differently.

Debrief Moves

- Knowledge question: Do categories describe reality or organise it?

- Who has authority to define terms?
- When does disagreement about words become disagreement about reality?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...